

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

Problem Set 2 - Solutions

(Exponential family of distributions, Completeness, UMVUE)

[1] Exponential Family Verification & Canonical Forms:

- (a) **Geometric distribution:** $p_\theta(x) = \theta^{x-1}(1 - \theta)$ for $x = 1, 2, \dots$

Derivation:

$$\begin{aligned} p_\theta(x) &= (1 - \theta)\theta^{x-1} \\ &= \frac{1 - \theta}{\theta} \theta^x \\ &= \exp\left(\ln\left(\frac{1 - \theta}{\theta}\right) + x \ln \theta\right) \end{aligned}$$

Canonical Form Identification:

- $h(x) = 1$
- **Sufficient Statistic:** $T(x) = x$
- **Natural Parameter:** $\eta = \ln \theta \implies \theta = e^\eta$
- **Log-partition Function:** Substitute $\theta = e^\eta$ into the constant term: $-\ln\left(\frac{1 - \theta}{\theta}\right) = -\ln(e^{-\eta} - 1)$

$$p_\eta(x) = \exp\{\eta x - \ln(e^{-\eta} - 1)^{-1}\}$$

- **Full Rank:** The range of $\theta \in (0, 1)$ implies $\eta \in (-\infty, 0)$, which is an open interval. **Yes, Full Rank.**

- (b) **Poisson distribution:** $p_\lambda(x) = e^{-\lambda} \frac{\lambda^x}{x!}$

Derivation:

$$\begin{aligned} p_\lambda(x) &= \frac{1}{x!} e^{-\lambda} \lambda^x \\ &= \frac{1}{x!} \exp(-\lambda + x \ln \lambda) \end{aligned}$$

Canonical Form Identification:

- $h(x) = 1/x!$
- **Sufficient Statistic:** $T(x) = x$
- **Natural Parameter:** $\eta = \ln \lambda \implies \lambda = e^\eta$
- **Log-partition Function:** $A(\eta) = \lambda = e^\eta$

$$p_\eta(x) = \frac{1}{x!} \exp\{\eta x - e^\eta\}$$

- **Full Rank:** $\lambda > 0 \implies \eta \in \mathbb{R}$ (Open set). **Yes, Full Rank.**

- (c) **t-distribution with v degrees of freedom:**

$$f(x) \propto \left(1 + \frac{x^2}{v}\right)^{-\frac{v+1}{2}} = \exp\left\{-\frac{v+1}{2} \ln\left(1 + \frac{x^2}{v}\right)\right\}$$

The term $\ln(1 + x^2/v)$ mixes the parameter v and the data x inside a non-linear function. It cannot be separated into a linear combination $\sum \eta_i T_i(x)$. **Conclusion: Not an Exponential Family.**

- (d) **Beta distribution:** $f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$

Derivation:

$$f(x) = \exp \{ (\alpha - 1) \ln x + (\beta - 1) \ln(1 - x) - \ln B(\alpha, \beta) \}$$

Canonical Form Identification:

- $h(x) = 1$
- **Sufficient Statistics:** $T(x) = (\ln x, \ln(1 - x))$
- **Natural Parameters:** $\eta_1 = \alpha - 1, \quad \eta_2 = \beta - 1$
- **Log-partition Function:** $A(\eta) = \ln B(\eta_1 + 1, \eta_2 + 1)$
- **Full Rank:** Parameter space $(\alpha, \beta) \in (0, \infty)^2$ maps to $\eta \in (-1, \infty)^2$, an open rectangle.
Yes, Full Rank.

- (e) **Binomial distribution:** $\text{Bin}(n, \theta)$ (n known)

Derivation:

$$\begin{aligned} p(x) &= \binom{n}{x} \theta^x (1 - \theta)^{n-x} \\ &= \binom{n}{x} (1 - \theta)^n \left(\frac{\theta}{1 - \theta} \right)^x \\ &= \binom{n}{x} \exp \left\{ x \ln \left(\frac{\theta}{1 - \theta} \right) + n \ln(1 - \theta) \right\} \end{aligned}$$

Canonical Form Identification:

- $h(x) = \binom{n}{x}$
- **Sufficient Statistic:** $T(x) = x$
- **Natural Parameter:** $\eta = \ln \left(\frac{\theta}{1 - \theta} \right)$ (Log-odds). Inverting: $\theta = \frac{e^\eta}{1 + e^\eta}$.
- **Log-partition Function:** Using $1 - \theta = \frac{1}{1 + e^\eta}$, term is $-n \ln(1 - \theta) = n \ln(1 + e^\eta)$.

$$p_\eta(x) = \binom{n}{x} \exp \{ \eta x - n \ln(1 + e^\eta) \}$$

- **Full Rank:** $\theta \in (0, 1) \implies \eta \in \mathbb{R}$. **Yes, Full Rank.**

- (f) **Normal distribution:** $N(\mu, \sigma^2)$

Derivation:

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right) \\ &= \exp \left(-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} - \frac{\mu^2}{2\sigma^2} - \ln(\sigma\sqrt{2\pi}) \right) \end{aligned}$$

Canonical Form Identification:

- **Sufficient Statistics:** $T(x) = (x, x^2)$
- **Natural Parameters:** $\eta_1 = \frac{\mu}{\sigma^2}, \quad \eta_2 = -\frac{1}{2\sigma^2}$
- **Full Rank:** The mapping covers $\mathbb{R} \times (-\infty, 0)$, an open set. **Yes, Full Rank.**

- (g) **Normal distribution with constrained parameters:** $N(\theta, \theta^2), \theta > 0$.

Using the natural parameters from (f):

$$\eta_1 = \frac{\theta}{\theta^2} = \frac{1}{\theta}, \quad \eta_2 = -\frac{1}{2\theta^2}$$

Rank Analysis: Notice that $\eta_2 = -\frac{1}{2}(\eta_1)^2$. The natural parameters lie on a parabola in the (η_1, η_2) plane. This curve does not contain any open rectangle in $\mathbb{R}^2$. **Conclusion: Curved Exponential Family (Not Full Rank).**

- (h) **Gamma distribution:** $f_{\alpha,\beta}(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$

Derivation:

$$f(x) = \exp \{ (\alpha - 1) \ln x - \beta x + \alpha \ln \beta - \ln \Gamma(\alpha) \}$$

Canonical Form Identification:

- **Sufficient Statistics:** $T(x) = (\ln x, x)$
- **Natural Parameters:** $\eta_1 = \alpha - 1, \quad \eta_2 = -\beta$
- **Full Rank:** $\eta_1 > -1, \eta_2 < 0$. Open rectangle. **Yes, Full Rank.**

- (i) **Distribution of $Y = \theta \ln X$:** Given $X \sim \text{Gamma}(\alpha, \beta)$ where α, θ are known constants. We are finding the family for Y indexed by β .

Transformation: $X = e^{Y/\theta}$. Jacobian $J = \frac{1}{\theta} e^{Y/\theta}$.

$$f_Y(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} (e^{y/\theta})^{\alpha-1} e^{-\beta e^{y/\theta}} \cdot \frac{1}{\theta} e^{y/\theta}$$

Collecting terms:

$$f_Y(y) = \underbrace{\frac{1}{\theta \Gamma(\alpha)}}_{h(y)} e^{\frac{\alpha y}{\theta}} \exp \left\{ -\beta e^{y/\theta} + \alpha \ln \beta \right\}$$

Canonical Form Identification:

- **Sufficient Statistic:** $T(y) = e^{y/\theta}$
- **Natural Parameter:** $\eta = -\beta$
- **Log-partition Function:** $A(\eta) = -\alpha \ln(-\eta)$
- **Full Rank:** $\beta > 0 \implies \eta \in (-\infty, 0)$. **Yes, Full Rank.**

- (j) **Power Series Distribution:** $p_\theta(x) = \frac{\gamma(x)\theta^x}{c(\theta)}$

Derivation:

$$p_\theta(x) = \gamma(x) \exp(x \ln \theta - \ln c(\theta))$$

Canonical Form Identification:

- $h(x) = \gamma(x)$
- **Sufficient Statistic:** $T(x) = x$
- **Natural Parameter:** $\eta = \ln \theta$
- **Full Rank:** Yes (assuming domain of θ is open).

[2] Joint Distribution & Minimal Canonical Form:

Given the PDF: $\exp\{-(\frac{x-\mu}{\sigma})^4 - \vartheta(\theta)\}$.

- (a) **Exponential Family Form:** Expand the term in the exponent:

$$-\frac{1}{\sigma^4}(x-\mu)^4 = -\frac{1}{\sigma^4}(x^4 - 4x^3\mu + 6x^2\mu^2 - 4x\mu^3 + \mu^4)$$

The joint distribution for a sample of size n is:

$$p_\theta(\underline{x}) \propto \exp \left\{ -\frac{1}{\sigma^4} \sum x_i^4 + \frac{4\mu}{\sigma^4} \sum x_i^3 - \frac{6\mu^2}{\sigma^4} \sum x_i^2 + \frac{4\mu^3}{\sigma^4} \sum x_i - nA(\theta) \right\}$$

This takes the form of an exponential family.

- (b) **Minimal Canonical Form:**

- **Minimal Sufficient Statistics:** $T(\underline{x}) = (\sum x_i^4, \sum x_i^3, \sum x_i^2, \sum x_i)$.
- **Natural Parameters:**

$$\eta_1 = -\frac{1}{\sigma^4}, \quad \eta_2 = \frac{4\mu}{\sigma^4}, \quad \eta_3 = -\frac{6\mu^2}{\sigma^4}, \quad \eta_4 = \frac{4\mu^3}{\sigma^4}$$

- **Rank Analysis:** We have 4 natural parameters $(\eta_1, \dots, \eta_4)$ completely determined by 2 parameters (μ, σ^2) . This implies functional dependencies between the η 's. The parameter space forms a 2-dimensional surface in $\mathbb{R}^4$.
- **Conclusion: Curved Exponential Family** (Not Full Rank).

[3] **Completeness of Hypergeometric Distribution:**

Let X_i indicate if the i -th item is defective. $T = \sum_{i=1}^n X_i$ is the number of defectives in the sample. $T \sim \text{Hypergeometric}(N, D, n)$, where $D \in \{0, 1, \dots, N\}$ is the unknown parameter.

PMF: $P_D(T = t) = \frac{\binom{D}{t} \binom{N-D}{n-t}}{\binom{N}{n}}$.

To prove completeness, we must show that $E_D[g(T)] = 0$ for all D implies $g(t) = 0$ for all t .

$$\sum_{t=0}^{\min(n,D)} g(t) \binom{D}{t} \binom{N-D}{n-t} = 0 \quad \forall D \in \{0, \dots, N\}$$

We solve this iteratively for $D = 0, 1, 2, \dots$:

- **D=0:** Only $t = 0$ is possible. $g(0) \binom{0}{0} \binom{N}{n} = 0 \implies g(0) = 0$.
- **D=1:** Possible values $t \in \{0, 1\}$. $g(0)P(T=0) + g(1)P(T=1) = 0$. Since $g(0) = 0$, we must have $g(1)P(T=1) = 0 \implies g(1) = 0$.
- **Induction:** Assume $g(t) = 0$ for $t = 0, \dots, k-1$. For $D = k$, the expectation sum includes terms up to $t = k$. All previous terms vanish, leaving $g(k)P(T=k) = 0 \implies g(k) = 0$.

Thus, $g(t) = 0$ for all possible t . **T is Complete.**

[4] **Completeness of Minimal Sufficient Statistics:**

(a) $U(\theta - 1/2, \theta + 1/2)$:

- **MSS:** $T = (X_{(1)}, X_{(n)})$.
- **Analysis:** The Range $R = X_{(n)} - X_{(1)}$ is an **ancillary statistic** (its distribution is independent of the location parameter θ). The expectation $E[R] = c$ is a known constant.
- **Counter-example:** Define $g(T) = R - c$. Then $E[g(T)] = 0$ for all θ , but $g(T)$ is not zero.
- **Conclusion: Not Complete.**

(b) $\text{Exp}(\theta, 1)$: PDF $f(x) = e^{-(x-\theta)} \mathbb{I}(x > \theta)$.

- **MSS:** $T = X_{(1)}$.
- **Analysis:** $E[g(T)] = \int_{\theta}^{\infty} g(t) n e^{-n(t-\theta)} dt = 0$. Divide by $n e^{n\theta}$ and differentiate with respect to θ (Leibniz rule):

$$\frac{d}{d\theta} \int_{\theta}^{\infty} g(t) e^{-nt} dt = -g(\theta) e^{-n\theta} = 0$$

This implies $g(\theta) = 0$ almost everywhere.

- **Conclusion: Complete.**

(c) $\text{Cauchy}(\theta, 1)$:

- **MSS:** Order statistics $(X_{(1)}, \dots, X_{(n)})$.
- **Analysis:** This is a location family. Any difference $X_{(i)} - X_{(j)}$ is an ancillary statistic. A bounded function of these differences can be constructed to have zero expectation.
- **Conclusion: Not Complete.**

[5] **Bivariate Normal Completeness:** $(X, Y) \sim N_2(0, 0, 1, 1, \theta)$.

- **MSS:** $T = (\sum X_i^2 + \sum Y_i^2, \sum X_i Y_i)$.

- **Analysis:** Marginally, $X_i \sim N(0, 1)$ and $Y_i \sim N(0, 1)$. Their second moments are $E[X_i^2] = 1$ and $E[Y_i^2] = 1$, regardless of the correlation θ .

$$E \left[\sum X_i^2 + \sum Y_i^2 \right] = 2n$$

Define $g(T) = (\sum X_i^2 + \sum Y_i^2) - 2n$. Then $E[g(T)] = 0$ for all θ , but $g(T) \neq 0$.

- **Conclusion: Not Complete.**

[6] **UMVUE for $\text{Bin}(1, \theta)$:** $T = \sum X_i$ is a Complete Sufficient Statistic (CSS) with distribution $\text{Bin}(n, \theta)$.

- (a) $g(\theta) = \theta$: Since $E[T] = n\theta$, the estimator T/n is unbiased. **UMVUE:** $\hat{g} = \frac{T}{n}$.
- (b) $g(\theta) = \theta^4$: We use factorial moments. $E[T(T-1)(T-2)(T-3)] = n(n-1)(n-2)(n-3)\theta^4$.
UMVUE: $\hat{g} = \frac{T(T-1)(T-2)(T-3)}{n(n-1)(n-2)(n-3)}$.
- (c) $g(\theta) = \theta(1-\theta)^2$: Let $\delta_0 = \mathbb{I}(X_1 = 1, X_2 = 0, X_3 = 0)$. $E[\delta_0] = \theta(1-\theta)^2$. Using Rao-Blackwellization $E[\delta_0|T = t]$:

$$\hat{g} = \frac{\binom{n-3}{t-1}}{\binom{n}{t}} = \frac{T(n-T)(n-T-1)}{n(n-1)(n-2)}$$

[7] **UMVUE for $U(0, \theta)$ of θ^2 :**

- $T = X_{(n)}$ is CSS.
- $E[T^k] = \frac{n}{n+k}\theta^k$.
- For $k = 2$, $E[T^2] = \frac{n}{n+2}\theta^2$.
- **UMVUE:** $\hat{g} = \frac{n+2}{n}T^2$.

[8] **UMVUE for $N(0, \theta)$ of θ^2 :** (where θ is variance)

- $T = \sum X_i^2$ is CSS. $E[T] = n\theta$.
- $\text{Var}(T) = 2n\theta^2 \implies E[T^2] = 2n\theta^2 + (n\theta)^2 = n(n+2)\theta^2$.
- **UMVUE:** $\hat{g} = \frac{T^2}{n(n+2)}$.

[9] **Examples of Sufficient vs. Complete Statistics:**

- (a) **A statistic that is sufficient but not complete:**

Example: Let $X_1, \dots, X_n$ be i.i.d. from a Uniform distribution $U(\theta, \theta + 1)$, where $\theta \in \mathbb{R}$.

Proof of Sufficiency: The joint PDF is:

$$f(\underline{x}|\theta) = \prod_{i=1}^n \mathbb{I}(\theta \leq x_i \leq \theta + 1) = \mathbb{I}(\theta \leq x_{(1)}) \cdot \mathbb{I}(x_{(n)} \leq \theta + 1)$$

where $x_{(1)} = \min(x_i)$ and $x_{(n)} = \max(x_i)$. By the Neyman-Fisher Factorization Theorem, $f(\underline{x}|\theta)$ depends on the data only through $T = (X_{(1)}, X_{(n)})$. Thus, T is a sufficient statistic.

Proof of Non-Completeness: Consider the Range statistic $R = X_{(n)} - X_{(1)}$. Since this is a location family ($X_i = \theta + \epsilon_i$ where $\epsilon_i \sim U(0, 1)$), the distribution of $R = \max(\epsilon_i) - \min(\epsilon_i)$ does not depend on θ . R is an *ancillary statistic*. The expected value $E[R] = \frac{n-1}{n+1}$ is a constant c . Define a function $g(T) = R - c = (X_{(n)} - X_{(1)}) - \frac{n-1}{n+1}$. Then:

$$E_\theta[g(T)] = E_\theta[R] - c = c - c = 0 \quad \forall \theta$$

However, $g(T)$ is a random variable and is not identically zero (almost everywhere). Therefore, T is **not complete**.

(b) **A statistic that is complete but not sufficient:**

Example: Let $X_1, \dots, X_n$ be i.i.d. from a Bernoulli distribution $Bern(\theta)$ with $n \geq 2$ and $\theta \in (0, 1)$. Let $T(\underline{X}) = X_1$ (the first observation only).

Proof of Completeness: We need to show that $E_\theta[g(T)] = 0 \forall \theta \implies g(t) = 0$. Since X_1 is Bernoulli, $g(t)$ takes values $g(0)$ and $g(1)$.

$$E_\theta[g(X_1)] = g(0)(1 - \theta) + g(1)\theta = 0$$

$$g(0) + \theta(g(1) - g(0)) = 0$$

For this linear equation in θ to hold for all $\theta \in (0, 1)$, both the intercept and the slope must be zero:

- Intercept: $g(0) = 0$
- Slope: $g(1) - g(0) = 0 \implies g(1) = 0$

Thus $g(t) \equiv 0$. Therefore, $T = X_1$ is **complete**.

Proof of Non-Sufficiency: A statistic is sufficient if the conditional distribution of the sample given the statistic is independent of θ . Consider the probability:

$$P(X_2 = 1 | T = X_1 = 1)$$

Since X_1, X_2 are independent:

$$P(X_2 = 1 | X_1 = 1) = P(X_2 = 1) = \theta$$

The conditional probability depends on θ . Therefore, $T = X_1$ is **not sufficient**.